THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL UNIT TEST SET 3

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The sum $1^3 + 2^3 + \cdots + 9^3$ equals: 1

(a) 729 **(b)** 90 **(c)** 2025 **(d)** 45

2. Rays OA, OB, OC, OD make equal angles around O (a full turn). The measure of each angle is: 1

(a) 82 **(b)** 62 **(c)** 72 **(d)** 162

3. On reversing and adding 187, the first sum is: 1

(a) 979 **(b)** 1067 **(c)** 968 **(d)** 957

4. A pictograph has symbols worth 12 each. To show 100, how many FULL symbols are needed (plus one part-symbol)? 1

(a) 10 **(b)** 8 **(c)** 9 **(d)** 7

5. The sum of the first 3 prime numbers after 10 is: 1

(a) 42 **(b)** 40 **(c)** 41 **(d)** 43

6. The 41st triangular number minus the 40th triangular number is: 1

(a) 42 **(b)** 41 **(c)** 40 **(d)** 430

7. A wheel has 12 equal spokes, so the angle between two neighbouring spokes is 30°. The angle between spoke 1 and spoke 6 (going one way round) is: 1

(a) 120 **(b)** 150 **(c)** 180 **(d)** 210

8. 68 is added to its reverse and the sum is 154. How many reverse-and-add steps in all are needed to reach a palindrome? 1

(a) 4 **(b)** 2 **(c)** 3 **(d)** 5

9. In a bar graph with scale 1 cm = 8 units, the value 60 is shown by a bar of height (cm): 1

(a) 52 **(b)** 7 **(c)** 7.5 **(d)** 8

10. The largest prime number less than 100 is: 1

(a) 96 **(b)** 97 **(c)** 99 **(d)** 98

11. At a party, 18 people each hug every other person exactly once. Number of hugs: 1

(a) 306 **(b)** 153 **(c)** 162 **(d)** 171

12. A clock shows 4 o'clock. After 5 hours, the smaller angle between the clock hands is: 1

(a) 120 **(b)** 270 **(c)** 90 **(d)** 150

13. **Assertion (A):** A straight angle measures 180°. 1
Reason (R): A right angle measures 90°.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** In the hailstone (Collatz) sequence, the number after 7 is 22. 1
Reason (R): If the number is odd, multiply by 3 and add 1; if it is even, divide by 2.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** In a bar graph all the bars are drawn with the same width. 1
Reason (R): The height (or length) of a bar shows the quantity.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the sum of odd numbers from 19 to 47. 2

17. 8 rays start from O, no two in the same direction, so that every pair makes one angle. Find the number of angles formed. 2

18. A number has 12 digits, all equal to 5. Find its digit sum, and say whether the number is divisible by 3 and whether it is divisible by 9. 2

19. The data shows the number of goals in 6 matches: 2, 3, 1, 4, 0, 5. Find the total number of goals and in how many matches more than 2 goals were scored. 2

20. Find the prime factors of 84 and the smallest number to multiply to make it divisible by 45. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the smallest number which when divided by 6, 8 and 12 leaves remainder 5 each time. 3

22. In a pattern the 1st term is 23, the 2nd is 30, and so on by adding 7. Which term of the pattern equals 331? Find also the term just before it. 3

23. At 5 o'clock the hands of a clock make a smaller and a larger angle. Find both and say which one is a reflex angle (if any). 3

24. Find a 4-digit palindrome of the form $abba$ with $a = 2$ and $b = 8$. Show that it is divisible by 11. 3

OR

The table shows two weeks of sales (units). Find the week with the larger total and by how much. 3

Day	Week 1	Week 2
Mon	14	15
Tue	20	20
Wed	13	14
Thu	17	13

25. Three traffic lights change every 30, 45 and 60 seconds. If they change together at 8:00 am, after how many seconds do they next change together and at what time? 3

OR

Sums of consecutive odd numbers give squares. Find the sum of the odd numbers from 1 to 49. Then find the sum of the odd numbers from 51 to 61. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Kavya has four digit cards: 5, 0, 8 and 3. She forms 4-digit numbers using every card exactly once. 5

(i) What is the largest number she can form?
(ii) What is the smallest 4-digit number she can form?
(iii) Find the difference between the largest and the smallest.
(iv) Find the smallest odd number she can form and explain why it is the smallest.

27. In a pattern of dot figures, figure 1 has 5 dots and each next figure has 9 more dots than the one before. How many dots are in figure 15? Which figure has 176 dots? 5

OR

At 4 o'clock, find 5
(a) the angle turned by the hour hand since 12 o'clock,
(b) the angle turned by the minute hand since 12 o'clock,
(c) the smaller angle between the hands,
(d) its type.

END OF QUESTION PAPER — Total: 50 Marks

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